CHAPTER 4
BEYOND CLASSICAL SEARCH

INTRODUCTION TO ARTIFICIAL INTELLIGENT (AI)

REV: 22.04.2024

So far we assumed that the environment is

- fully observable,
- deterministic

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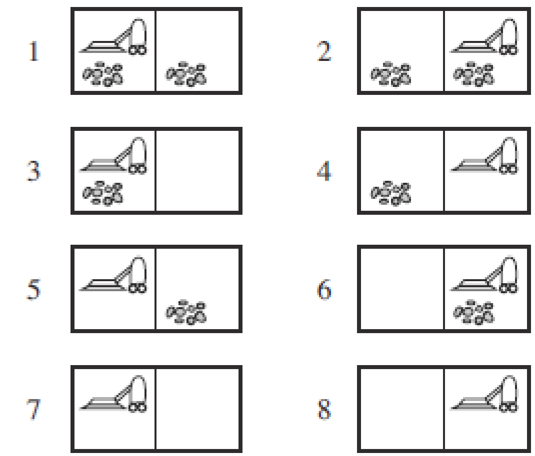
Now we investigate non-deterministic environment, unobservable environment and partially observable environment, respectively.

Searching with Nondeterministic Actions



The eight possible states of the vacuum world; states 7 and 8 are goal states.

Consider an erratic vacuum world, the Suck action works as follows:

- when applied to a dirty square the action clean, it cleans this square and sometimes cleans an adjacent square, too.
- when applied to a clean square the action clean, it sometimes deposits dirt on this square.

Searching with Nondeterministic Actions

Definition for the Nondeterministic Search Problems

The components such as states, initial states, actions, etc. remains same. However, transition model should be modified.

- **Transition model:**

In classical search problem, a single state s' is produced from the current state s and the action a:

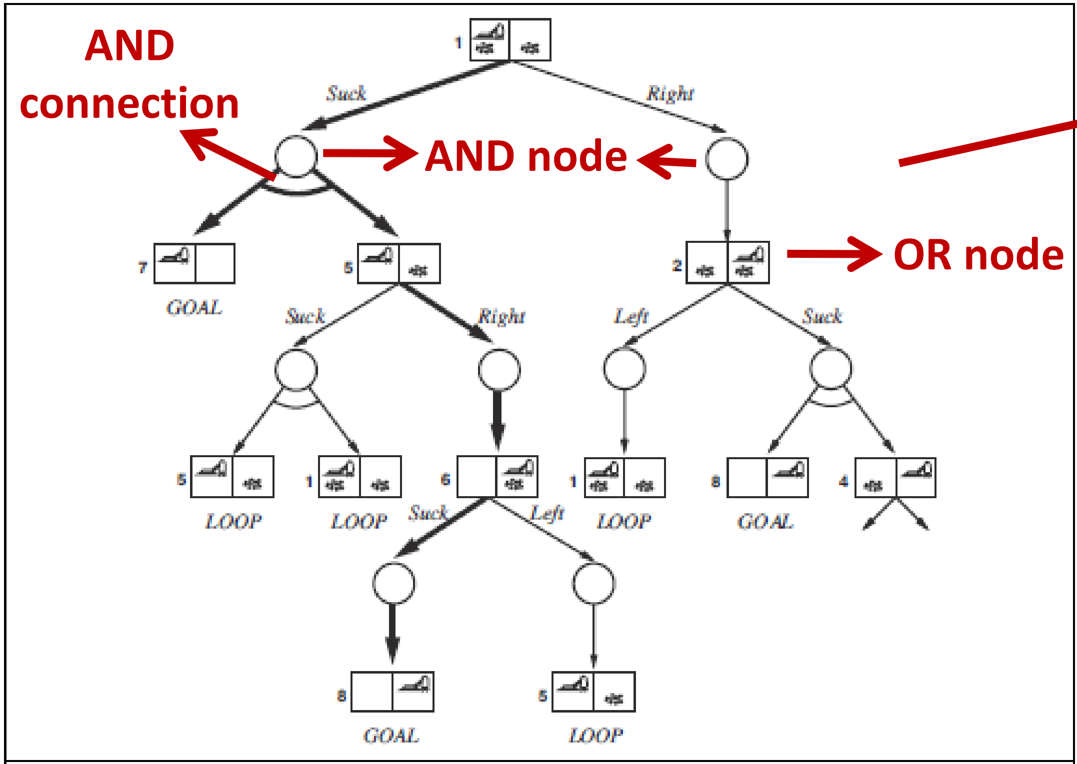
$$s' = RESULT(s, a)$$

For deterministic case, the transition model can produce more than one states:

$$\{s'_1, s'_2, \dots\} = RESULTS(s, a)$$

Therefore, AND nodes are included in the search tree.

Searching with Nondeterministic Actions



The first two levels of the search tree for the erratic vacuum world. State nodes are OR nodes where some action must be chosen. At the AND nodes, shown as circles, every outcome must be handled, as indicated by the arc linking the outgoing branches. The solution found is shown in bold lines.

An **AND–OR** Search Tree

A search tree for the Erratic Vacuum World.

Searching with Nondeterministic Actions

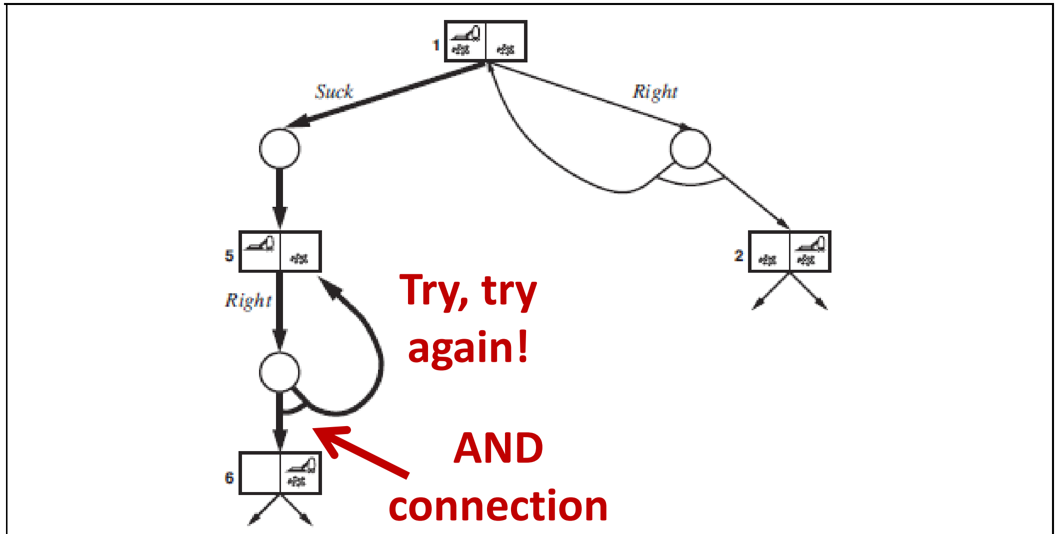
For nondeterministic case, the solution to a problem is not a sequence but a strategy that specifies what to do depending on what percepts are received.

Thus, solutions for nondeterministic problems contain nested if–then–else statements such as:

[*Suck*, if State =5 then [*Right*, *Suck*] else [ ]]

This means that the solutions are trees rather than sequences. More clearly, a solution for an AND–OR search problem is a subtree that
(1) has a goal node at every leaf,
(2) specifies one action at each of its OR nodes, and
(3) includes every outcome branch at each of its AND nodes.

Searching with Nondeterministic Actions



Part of the search graph for the slippery vacuum world, where we have shown (some) cycles explicitly. All solutions for this problem are cyclic plans because there is no way to move reliably.

Consider the slippery vacuum world, which is identical to the ordinary (non-erratic) vacuum world except that movement actions sometimes fail, leaving the agent in the same location.

[*Suck*, while State =5 do [*Right*], [*Suck*]]

Example: Investigate the problem given in 'AI_8_Example_undeterministic_environment'

Searching with No Observations

Is it possible make a search without a sensor?

Depends on the problem

- for 8-puzzle?

or

- for a sensorless vacuum cleaner?

Searching with No Observations

Is it possible make a search without a sensor?

Depends on the problem

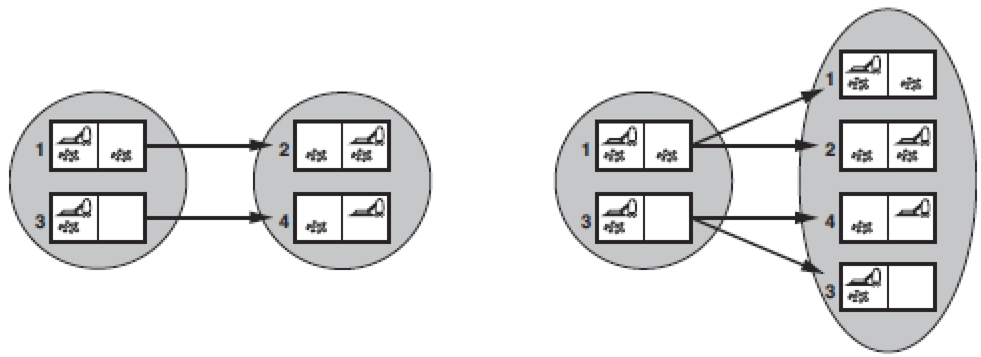
- for 8-puzzle? → IMPOSSİBLE!

or

- for a sensorless vacuum cleaner? → YES!

Searching with No Observations

A belief state



(a) (b)

(a) Predicting the next belief state for the sensorless vacuum world with a deterministic action, *Right*. (b) Prediction for the same belief state and action in the slippery version of the sensorless vacuum world.

When the agent has no sensor (in the case of an unobservable environment), we search in the space of belief states rather than classical single states (physical states). A belief state is a set of possible states where the agent might be.

Searching with No Observations

In the case of an unobservable environment, the solution (if any) is always a sequence of actions. This is true even if the environment is nondeterministic.

Let P be physical problem that is defined by $ACTIONS_P, RESULT_P$, $GOAL\text{-}TEST_P$ and $STEP\text{-}COST_P$. Then the corresponding sensorless problem can be defined as follows:

- **Belief states:** The entire belief-state space contains every possible set of physical states. If P has N states, then the sensorless problem has 2^N belief states, although many may be unreachable from the initial state.
- **Initial state:** The set of all states in P.
- **Actions:** Suppose the agent is in belief state $b = \{s_1, s_2, s_3, \ldots\}$, but $ACTIONS_P(s_i) \neq ACTIONS_P(s_j)$; then the agent is unsure of which actions are legal.

If illegal actions have no effect on the environment ⟹ $ACTIONS(b) = \bigcup_{s \in b} ACTIONS_P(s)$

If an illegal action might be the end of the world ⟹ $ACTIONS(b) = \bigcap_{s \in b} ACTIONS_P(s)$

Searching with No Observations

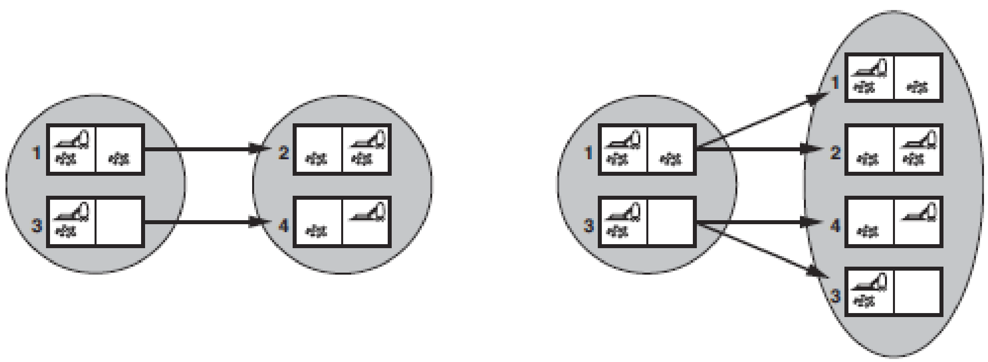
- **Transition model:** For deterministic actions, the set of states that might be reached is:

$$b' = PREDICT(b, a) = \{s' : s' = RESULT_P(s, a), s \in b\}$$

With deterministic actions, b' is never larger than b. With nondeterminism, we have

$$b' = PREDICT(b, a) = \{s' : s' = RESULTS_P(s, a), s \in b\}$$

which may be larger than b.



(a) (b)

(a) Predicting the next belief state for the sensorless vacuum world with a deterministic action, *Right*. (b) Prediction for the same belief state and action in the slippery version of the sensorless vacuum world.

Searching with No Observations

- **Goal test:** a belief state satisfies the goal only if all the physical states in it satisfy $GOAL-TEST_P$.

- **Path cost:** If the same action can have different costs in different states, then the cost of taking an action in a given belief state could be one of several values. If we assume that the cost of an action is the same in all states and so can be transferred directly from the underlying physical problem.

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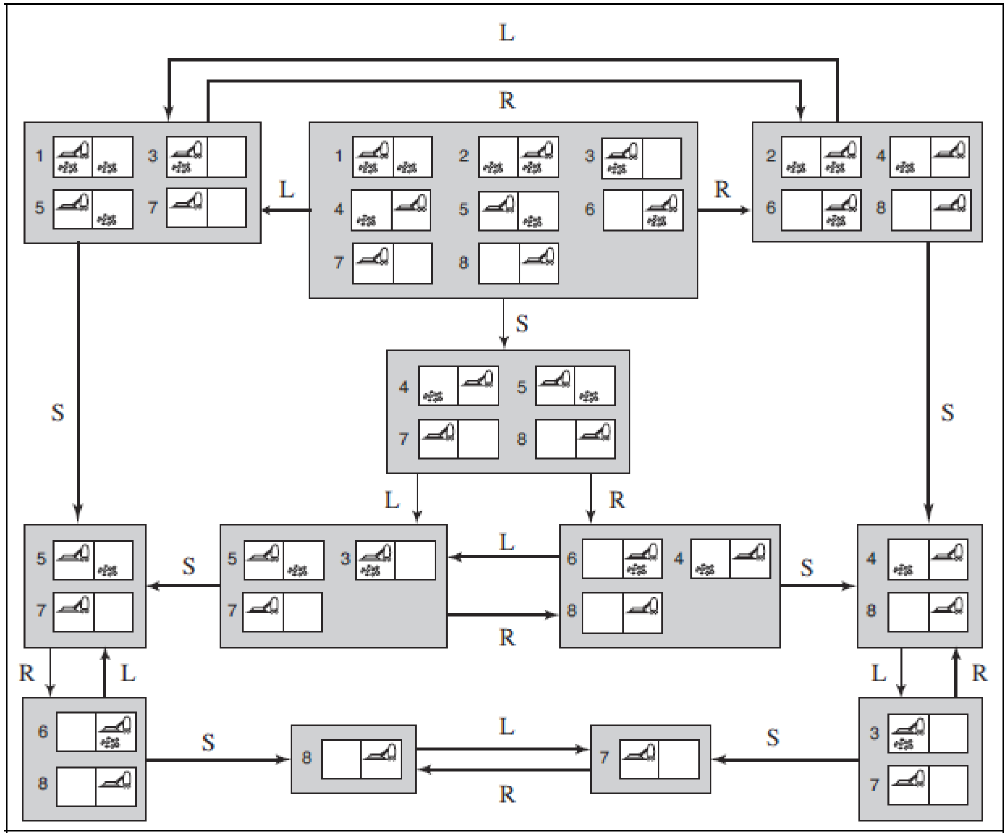
We can perform a search tree algorithm in a space of belief states

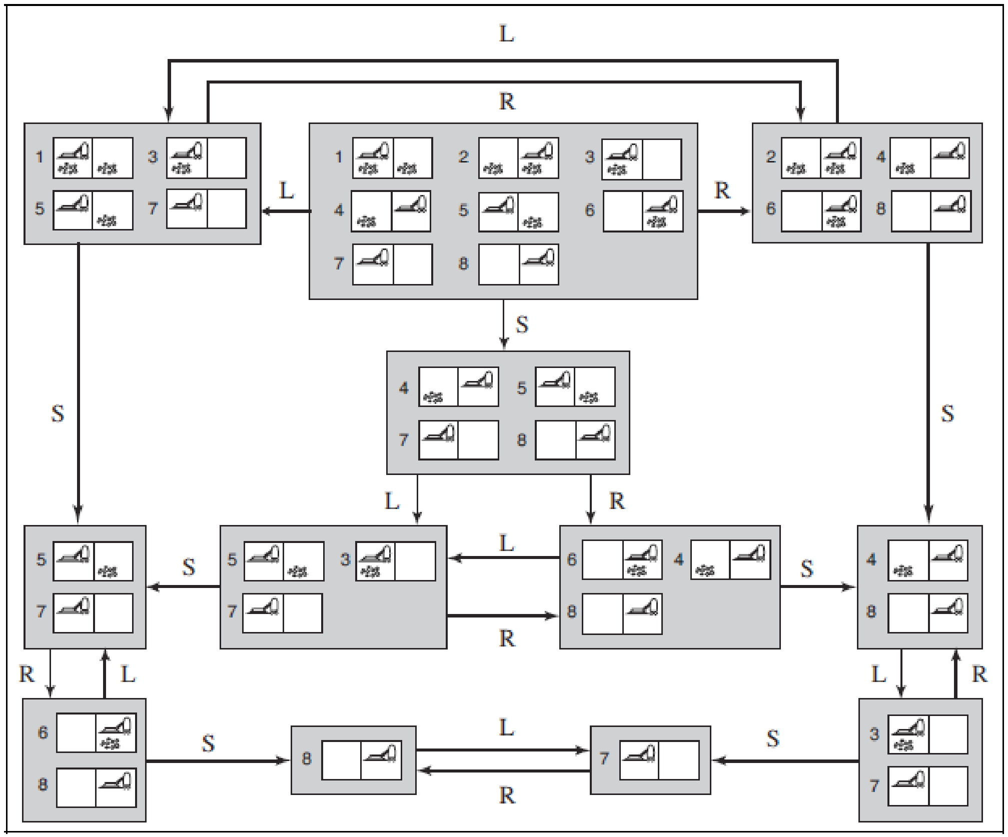
Searching with No Observations



The **belief-state** space for the **deterministic, sensorless vacuum world**. There are only 12 reachable belief states out of $2^8 = 256$ possible belief states.

Each shaded box corresponds to a single belief state. At any given point, the agent is in a particular belief state but does not know which physical state it is in. Self-loops are omitted for simplicity.

Searching with No Observations

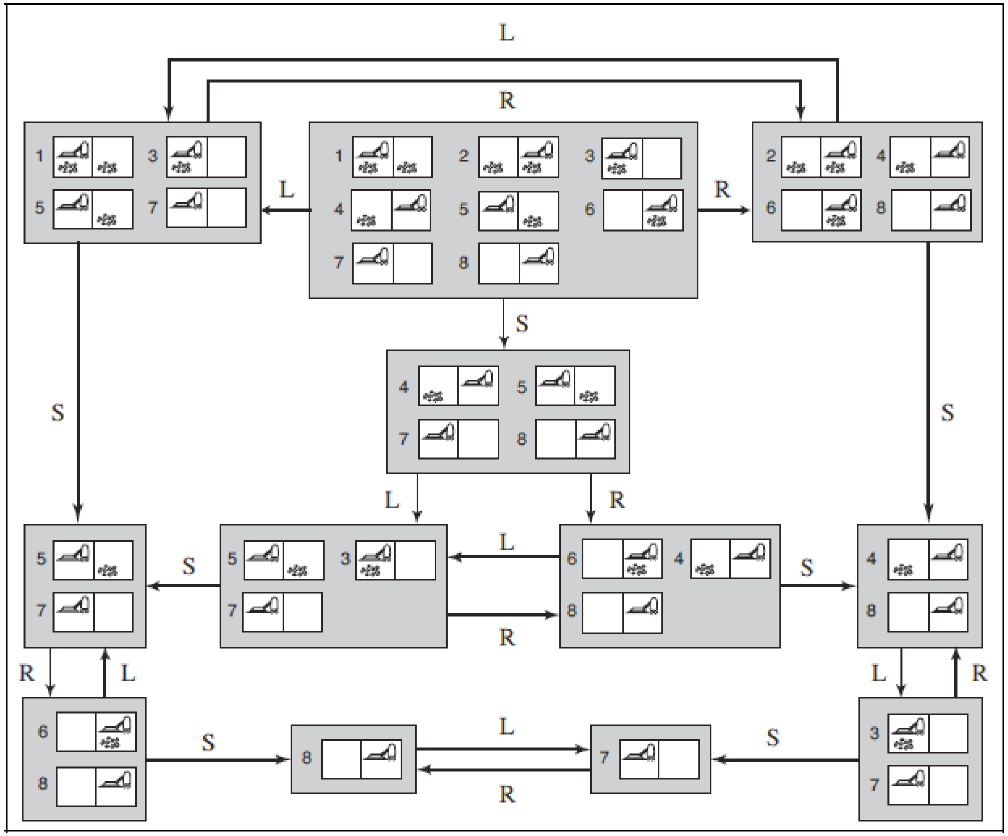


Consider the problem for a search tree!

Or an alternative approach

The incremental belief-state search algorithms that build up the solution one physical state at a time, namely…

Searching with No Observations



In the sensorless vacuum world, the initial belief state is {1, 2, 3, 4, 5, 6, 7, 8}, and we have to find an action sequence that works in all 8 states. We can do this by first finding a solution that works for state 1; then we check if it works for state 2; if not, go back and find a different solution for state 1, and so on. The incremental belief-state search has to find one solution that works for all the states.

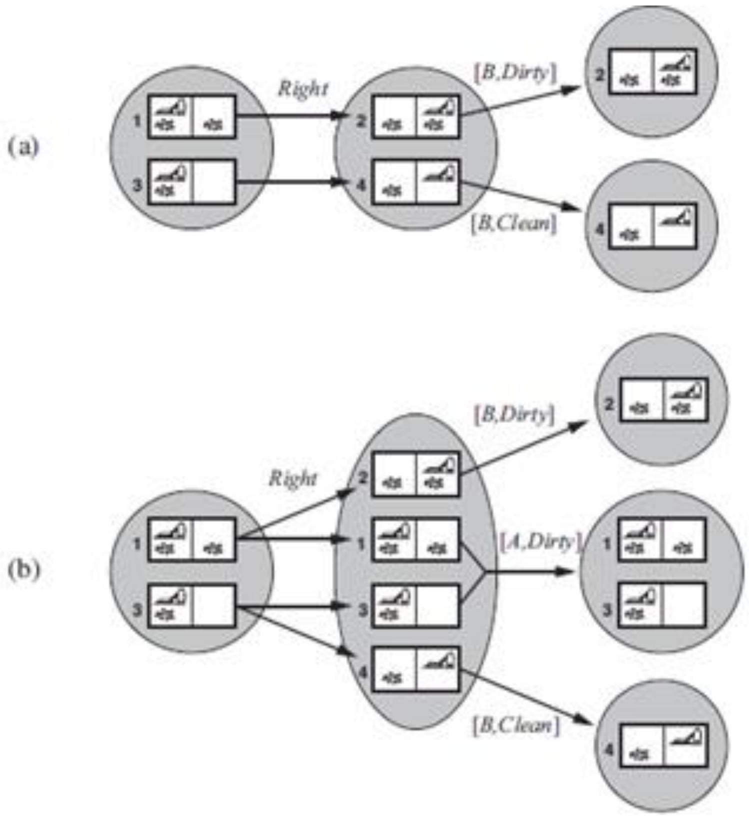
Searching with Partial Observations

The sensorless 8-puzzle is impossible to solve. But, is it possible make a search for 8-puzzle if just one square is visible?

Yes

The solution involves moving each tile in turn into the visible square and then keeping track of its location.

Searching with Partial Observations



The local-sensing vacuum world to be one in which the agent has a position sensor and a local dirt sensor but has no sensor capable of detecting dirt in other squares.

Fig (a) shows deterministic world while Fig (b) shows slippery world.

Searching with Partial Observations

When the environment is partially observable, the transition model becomes complicated according to the sensorless (unobservable) case. While the transition model is composed of only the prediction stage in the sensorless case, it is composed of 3 stages, namely prediction, observation prediction and update, in the partially observable case.

- **Transition model:** Let b be the current belief state and a be the action to be performed.

$b' = PREDICT(b, a)$ → b' is the predicted belief states given the action a.

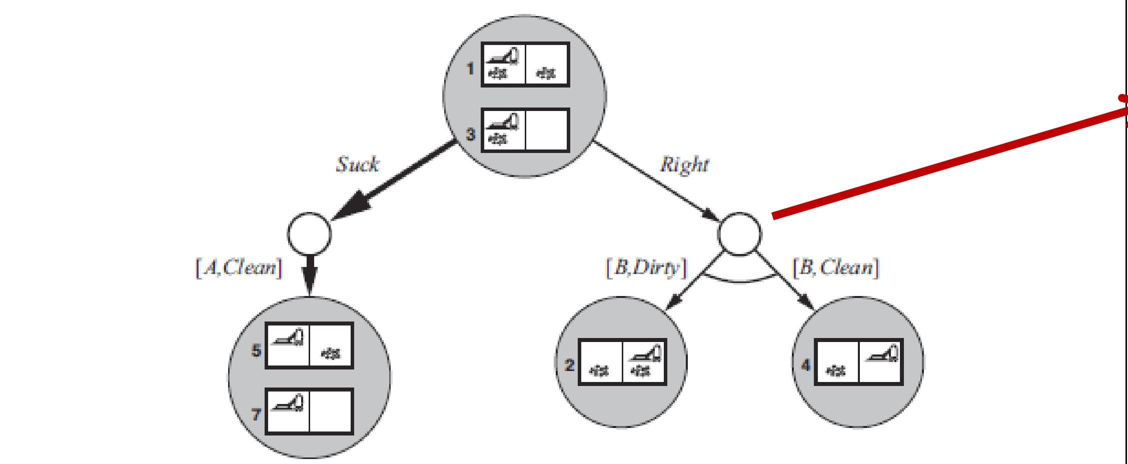
$O = POSSIBLE\_PERCEPTS(b') = \{o : o = PERCEPT(s), s \in b'\}$

The set of percepts o that could be observed in the predicted belief state.

$b_o = UPDATE(b', O) = \{s : o = PERCEPT(s), s \in b'\}$

For each possible percept, the belief state that would result from the percept.

Searching with Partial Observations



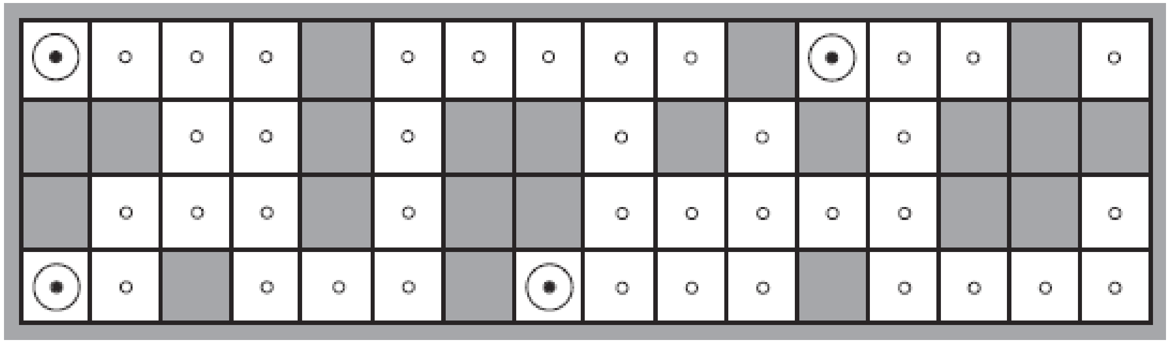
The first level of the AND–OR search tree for a problem in the local-sensing vacuum world; *Suck* is the first step of the solution.

Since partial observation creates undeterminism and the agent can decrease undeterminism by partially observing for the future states, **AND** nodes are used.

The local-sensing vacuum world (for deterministic case) to be one in which the agent has a position sensor and a local dirt sensor but has no sensor capable of detecting dirt in other squares.

SOLUTION STRATEGY → **[*Suck*, Right, if BeliefState =6 then [*Suck*] else [ ]]**

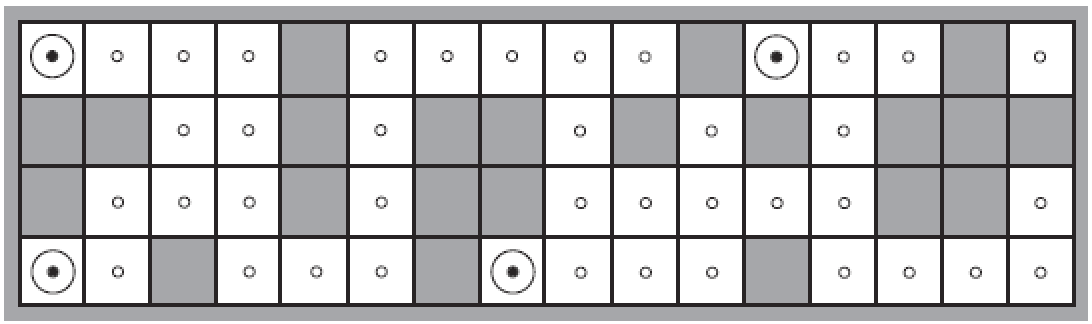
Searching with Partial Observations (Localization)



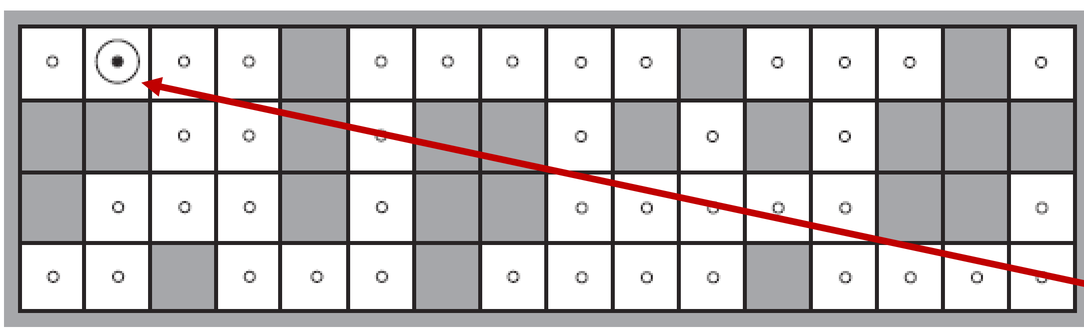
(a) Possible locations of robot after $E_1 = NSW$

The robot is placed in the maze-like environment. It is equipped with four sonar sensors that tell whether there is an obstacle in each of the four compass directions. We assume that the robot has a correct map of the environment. But unfortunately the robot's navigational system is broken, so when it executes a Move action, it moves randomly to one of the adjacent squares. The robot's task is to determine its current location.

Searching with Partial Observations (Localization)



(a) Possible locations of robot after E_1 = NSW



(b) Possible locations of robot After E_1 = NSW, E_2 = NS

The First Observation:
North, South and West are closed.

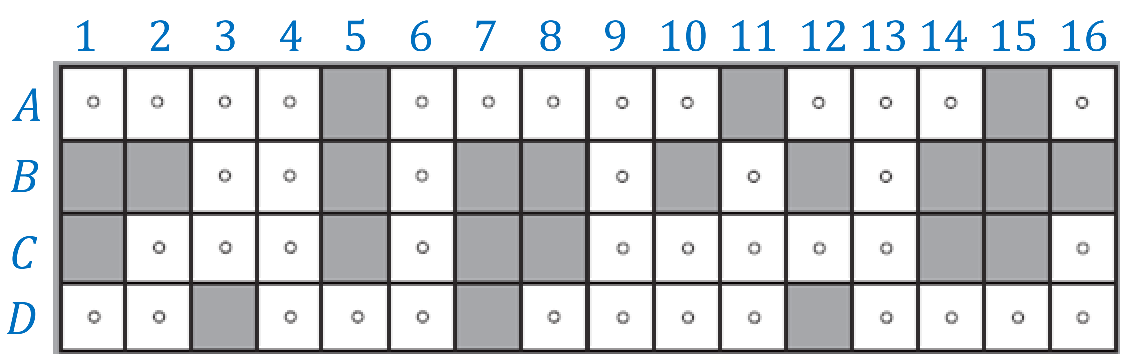
Move East

The Second Observation:
North and South are closed.

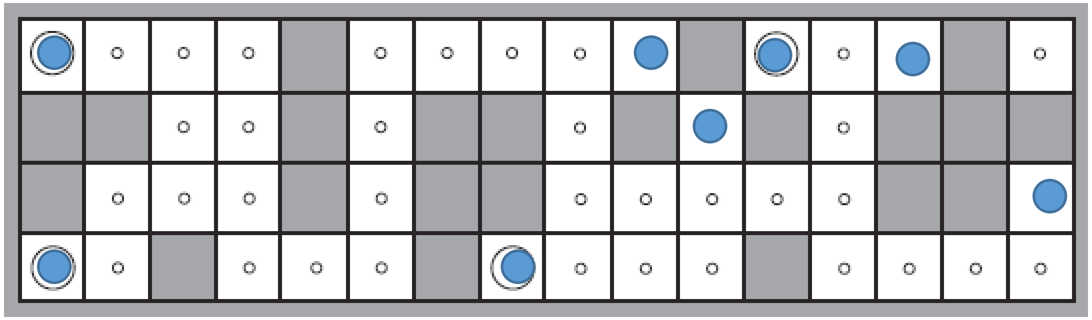
The exact location is found.

Searching with Partial Observations (Localization)

Example: If the first percept of the robot defined previously is that South and North are closed, apply depth-first search algorithm and draw the search tree.



Searching with Partial Observations (Localization)



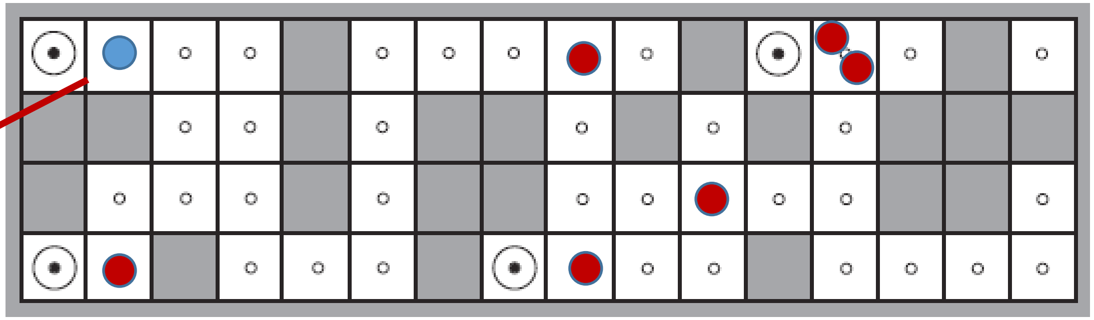
(a) Possible locations of robot after $E_1 = NSW$

Consider the same problem if the robot's compass is also broken?

The First Observation : Three directions are closed.
Move One Square

Searching with Partial Observations (Localization)

The exact location is found.



(a) Possible locations of robot after $E_1 = NSW$

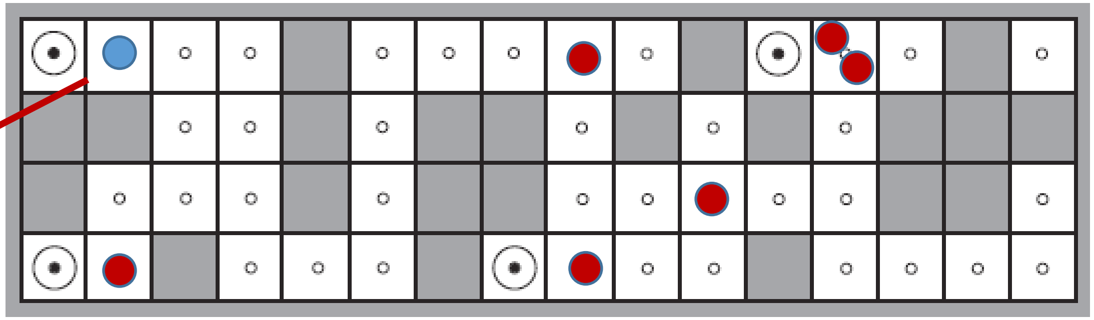
The First Observation : Three directions are closed.

Move One Square

The Second Observation : Only the previous moving and its opposite side are open.

Searching with Partial Observations (Localization)

The exact location is found.



(a) Possible locations of robot after E_1 = NSW

For the all maps, does that approach yield result?

No.

For the some symmetric mazes, it does not work.

Main Reference

Stuart Russell, Peter Norvig; “Artificial Intelligence A Modern Approach”, Pearson, 3rd Edition, 2013.